



Genome-wide characterization of the ankyrin repeats gene family under salt stress in soybean



Dayong Zhang^{a,*}, Qun Wan^{a,1}, Xiaolan He^a, Lihua Ning^a, Yihong Huang^a, Zhaolong Xu^a, Jia Liu^a, Hongbo Shao^{a,b,**}

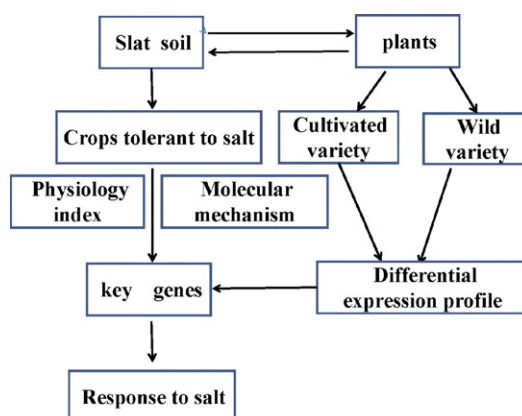
^a Provincial Key Laboratory of Agrobiolgy, Institute of Biotechnology, Jiangsu Academy of Agricultural Sciences, Nanjing 210014, China

^b Key Laboratory of Coastal Biology & Bioresources Utilization, Yantai Institute of Coastal Zone Research (YIC), Chinese Academy of Sciences (CAS), Yantai 264003, China

HIGHLIGHTS

- The genome-wide ANK family genes were identified in soybean.
- The GmANKs genes related to salt stress were identified.
- The response to salt stress of GmANK6 was performed.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 13 May 2016

Received in revised form 11 June 2016

Accepted 12 June 2016

Available online 19 June 2016

Keywords:

GmANKs

Bioinformatics analyses

GmANK39-like

Salt stress

Expression

Subcellular localization

ABSTRACT

Ankyrin repeats (ANK) gene family are common in diverse organisms and play important roles in cell growth, development and response to environmental stresses. Recently, genome-wide identification and evolutionary analyses of the ANK gene family have been carried out in *Arabidopsis*, rice and maize. However, little is known about the ANK genes in the whole soybean genome. In this study, we described the identification and structural characterization of 162ANK genes in soybean (GmANK). Then, comprehensive bioinformatics analyses of GmANK genes family were performed including gene locus, phylogenetic, domain composition analysis, chromosomal localization and expression profiling. Domain composition analyses showed that GmANK proteins formed eleven subfamilies in soybean. In silico expression analysis of these GmANK genes demonstrated that GmANK genes show a diverse/various expression pattern, suggesting that functional diversification of GmANK genes family. Based on digital gene expression profile (DGEP) data between cultivated soybean and wild type under salt treatment, some GmANKs related to salt/drought response were investigated. Moreover, the expression pattern and subcellular localization of GmANK6 were performed. The results will provide important clues to explore ANK genes expression and function in future studies in soybean.

© 2016 Elsevier B.V. All rights reserved.

* Corresponding author.

** Corresponding author at: Provincial Key Laboratory of Agrobiolgy, Institute of Biotechnology, Jiangsu Academy of Agricultural Sciences, Nanjing 210014, China.

E-mail addresses: cotton.z@126.com (D. Zhang), shaohongbuchu@126.com (H. Shao).

¹ These authors contributed equally to this work.

1. Introduction

Soil salinization, which affects an estimated 1 to 3 million ha in the enlarged EU and >3.5 million ha in China, is one of the serious problems restricting agricultural development in the world (<http://www.fao.org/soils-portal/soil-management/management-of-some-problemsoils/salt-affected-soils/more-information-on-salt-affected-soils/en/>). Most crop plants are sensitive to excess salt concentration in soil (Xiong et al., 2002), it is of great importance to insight into the mechanisms underlying plant response to salinity stress, especially for plant-salt soil interaction. Cultivating the novel varieties with high tolerance to salt is one of the most effective methods, therefore, mining the key genes controlling crop tolerance to salt becomes more and more important for modern agriculture, especially via high throughput technologies.

Ankyrin-repeat domains contain a 33-amino acid motif that is widely present in prokaryotes, eukaryotes and some viruses, and they play an important role in mediating protein-protein interactions (Sedgwick and Smerdon, 1999). In plants, proteins containing ankyrin repeats are involved in diverse cellular functions, including cell cycle regulation, cytoskeleton interactions, signal transduction, ion transport (Sedgwick and Smerdon, 1999) and organ development, including leaf morphogenesis (Ha et al., 2004) and chloroplast biogenesis (Bae et al., 2008). Additionally, they play important roles in plant responses to biotic and abiotic stresses (Yan et al., 2002; Lu et al., 2005; Seong et al., 2007; Sharma and Pandey, 2015). In recent years, genome-wide analysis of ankyrin-repeat proteins in maize (Jiang et al., 2013) has been reported. Sakamoto et al. (2008) identified *itn1*, a novel gene with an ankyrin-repeat motif in *Arabidopsis thaliana* that affects the ABA-mediated production of reactive oxygen species and is involved in salt-stress tolerance, suggesting that *itn1* is involved in the ABA-dependent salinity stress pathway. Ito et al. (1999) characterized a protein containing ankyrin repeats named Ankhzn, which localized to the endosomal membrane and was involved in vesicle or protein transport, and it bound to a zinc finger motif. In general, ankyrin-repeat domains mediate protein-protein interactions and act as molecular scaffolds for protein recognition (Sedgwick and Smerdon, 1999).

Soybean (*Glycine max* L.) is one of the important crops providing plant protein and oil. Here, we identified 162 GmANK genes in soybean by database searches, which genes were classified by protein domains. We analyzed the gene locus and Chromosome (Chr) localization of the protein length, number of ANK domain, molecular weight (MW), isoelectric points (pI), expression patterns and phylogenetic relationship of the ANK genes in soybean. We also surveyed the expression patterns of *GmANK39-like* (*GmANK6*) gene in the developmental stages of soybean and its responses to abiotic stress and subcellular localization.

2. Materials and methods

2.1. Obtaining of protein sequences containing ANK domain in soybean genome

To identify soybean proteins containing Ankyrins domain at the genome level, all protein-coding genes were downloaded from the latest version of the soybean genome (Wm82.a2.v1) in Phytozome v.10.0. Hidden Markov models (HMMs) of the 'typical' Ankyrin clan (PF00023) was to search for putative PKs using HMMER v.3.0 with an E-value cut-off of <1.0.275 sequences containing the Ankyrin domain were identified. Only the longest variant of each gene was retained and all other redundant sequences were deleted. Domain composition was searched by SMART and PFAM database, Molecular Weight and isoelectric points was calculated with DNASTar software.

To further reveal the conserved motif of the GmANKs protein, the motif investigation software Multiple EM for Motif Elicitation (MEME) (Bailey et al., 2006) (version 4.0; <http://meme.sdsc.edu/mere4/cgi-bin/meme.cgi>) was carried out. The parameters of this analysis were as follows: optimum motif width ranged from 6 to 50; the maximum

number of motifs for identification was 20, 3 motifs should find and all other parameters were defaulted.

2.2. In silico subcellular predicted localization and expression analyses for GmANKs

The protein subcellular localization was predicted using the online tool WoLF PSORT available at <http://wolfpsort.org/>, and the expression data for GmANKs in different tissues was downloaded from the <http://soybase.org>, analyzed in silico and displayed using Java-Treewiew software.

2.3. Plant material

The *Glycine max* var. Williams 82 variety was used to grow seedlings and extract total RNA for *GmANK-39* gene cloning, tissue expression and induced expression analysis experiments.

2.4. Gene cloning and sequence analysis

The *GmANK39-like* gene primers (forward primer:ATGCTTGCCA TCGGGAAGGCCACC, reverse primer:TCAATGAAAGAGCAAAAAACATA) were designed according to the full-length coding sequence and used to cloned the genes from soybean root tissue cDNA using RT-PCR (Reverse transcriptase-polymerase chain reaction).

2.5. Phylogenetic analyses

The Neighbour Joining (NJ) tree of GmANKs from soybean and other plants was performed using MEGA5.0 software using neighbour-joining (NJ) and maximum-likelihood (ML) methods (Tamura et al., 2011).

2.6. qPCR analysis

The soybean root, stem, leaf at young seedling stage, and flower, pod were harvested and frozen in liquid nitrogen for RNA extraction. The soybean roots were collected from plants treated with PEG6000 for 0 h, 2 h, 4 h and 8 h or with 200 mM NaCl for 0 h, 1 h, 2 h, 4 h, 8 h, 12 h, 24 h, 48 h and 72 h. Total RNA was extracted from soybean samples according to the TRIZOL Kit (Invitrogen) manual. Single-stranded cDNA was synthesized using 2 µg total RNA and Oligod(T)18 primer with the Takara RT-PCR system in a total volume of 25 µl according to the manufacturer's protocol (TaKaRa Bio Inc.) and used for qRT-PR analysis. qPCR analysis was performed as described previously (Zhang et al., 2013). The gene primers used were forward primer:ATGCTTGCCATCGGGAAGGCCACC, reverse primer:TCAATGAAA GAGCAAAAAACATA, the internal control gene was *GmActin2* forward primer:CGGTGGTCTATCTTGGCATC, reverse primer:GTCTTTCGC TCAATAACCTA. The assays were repeated three times.

2.7. Subcellular localization

PCR-generated *HindIII-BamHI* fragment containing the open reading frame of *GmANK39-like* without stop codon (Forward primer: cgcGTCGACATGCTTGCCATCGGG, reverse primer:cgcCCATGGATGAA AGAGCAAAAAACAT) was subcloned in-frame upstream of the GFP gene in the pJIT166GFP plasmid. The construct was validated by sequencing.

Arabidopsis leaf protoplasts were isolated according to Yoo et al. (2007). The resulting fusion construct (p35S::GmANK39-like-GFP) or empty control vector (p35S::GFP) were introduced into *Arabidopsis* protoplasts by the PEG4000-mediated method (Abel and Theologis, 1994). After 24 h at room temperature, GFP signal was detected by confocal fluorescence microscopy (Zeiss, LSM510 Meta, Carl Zeiss AG).

Table 1
The ANK gene family in soybean.

Gene locus	Chr	Gene name	No. ANK	Length (aa)	MW (KD)	pI	BLAST2GO results	SL
Glyma01G052600	1	GmANK1	1	428	48.02	7.16	Mitogen-activated protein kinase kinase kinase Raf17.1	cysk
Glyma01G055500	1	GmANK2	5	276	29.99	5.27	Ankyrin repeat domain-containing protein	cyto
Glyma01G149900	1	GmANK3	3	252	26.87	6.18	Ankyrin repeat domain-containing protein	nucl
Glyma01G150400	1	GmANK4	3	173	19.22	7.81	Ankyrin repeat domain-containing protein	cyto
Glyma01G161900	1	GmANK5	5	443	50.61	7.58	Ankyrin repeat domain-containing protein	chlo
Glyma02G033700	2	GmANK6	2	143	15.86	6.9	Ankyrin repeat domain-containing protein	cyto
Glyma02G113900	2	GmANK7	5	244	26.03	4.96	Ankyrin repeat family protein	cyto
Glyma02G151200	2	GmANK8	5	456	49.13	7.49	Ankyrin repeat domain-containing protein	nucl
Glyma02G188700	2	GmANK9	2	477	51.97	6.15	Ankyrin repeat family protein	ER
Glyma02G227000	2	GmANK10	2	696	75.83	5.9	Zinc finger CCCH domain-containing protein	nucl
Glyma02G243400	2	GmANK11	5	850	97.4	6.28	Potassium channel SKOR	plas
Glyma02G263600	2	GmANK12	4	352	38.4	6.66	Ankyrin repeat domain-containing protein	cyto
Glyma02G288300	2	GmANK13	3	455	51.49	7.15	Protein kinase family protein	cyto
Glyma02G296600	2	GmANK14	2	658	71.84	7.19	Zinc finger CCCH domain-containing protein	nucl
Glyma03G128600	3	GmANK15	2	488	53.34	6.44	BTB/POZ ankyrin repeat protein	cyto
Glyma03G138600	3	GmANK16	3	681	74.91	5.89	Zinc finger CCCH domain-containing protein	nucl
Glyma03G170400	3	GmANK17	5	628	69.54	5.44	Ankyrin repeat family protein	plas
Glyma03G170500	3	GmANK18	4	574	63.81	6.57	Ankyrin repeat family protein	plas
Glyma03G174400	3	GmANK19	7	579	63.24	9.6	Ankyrin repeat-containing protein	plas
Glyma03G174500	3	GmANK20	8	522	56.97	8.8	Ankyrin repeat-containing protein	plas
Glyma03G247600	3	GmANK21	5	465	49.69	5.1	Ankyrin repeat family protein	chlo
Glyma03G264500	3	GmANK22	8	567	62.44	9.4	Ankyrin repeats-containing protein	plas
Glyma04G044500	4	GmANK23	3	372	41.28	5.45	Arginine N-methyltransferase	cyto
Glyma04G069400	4	GmANK24	5	879	99.05	6.43	Potassium channel	plas
Glyma04G116500	4	GmANK25	3	351	37.633	4.39	Ankyrin-repeat protein	mito
Glyma04G134700	4	GmANK26	9	963	104.15	5.86	E3 ubiquitin-protein ligase KEG	chlo
Glyma04G200800	4	GmANK27	2	786	87.9	3.69	ADP-ribosylation factor GTPase-activating protein	nucl
Glyma04G233600	4	GmANK28	3	347	37.88	4.32	ACYL-COA binding family protein	ER
Glyma05G010600	5	GmANK29	4	894	100.16	6.91	Potassium channel	plas
Glyma05G088400	5	GmANK30	9	539	58.1	8.6	NA	chlo
Glyma05G117000	5	GmANK31	1	1089	121.94	5.67	Calmodulin-binding transcription activator	nucl
Glyma05G124700	5	GmANK32	9	1032	113.99	7.45	NA	plas
Glyma05G145200	5	GmANK33	9	675	74.51	7.79	Serine/threonine protein phosphatase 6 regulatory ankyrin repeat subunit C	chlo
Glyma05G148300	5	GmANK34	2	984	109.31	5.47	Calmodulin-binding family protein	nucl
Glyma05G167600	5	GmANK35	2	476	53.66	9.05	Ankyrin-kinase	mito
Glyma05G178200	5	GmANK36	2	1123	126.99	5.99	Ethylene-responsive calmodulin-binding family protein	nucl
Glyma05G200700	5	GmANK37	2	801	90.32	7.05	ADP-ribosylation factor GTPase-activating protein	nucl
Glyma05G238600	5	GmANK38	5	531	58.73	7.74	Ankyrin repeat-containing protein	chlo
Glyma05G247500	5	GmANK39	6	780	89.15	6.65	Outward rectifying potassium channel	plas
Glyma06G045000	6	GmANK40	3	329	36.33	4.64	Arginine N-methyltransferase	nucl
Glyma06G071100	6	GmANK41	4	880	99.49	6.46	Inward rectifying shaker-like K ⁺ channel	plas
Glyma06G131200	6	GmANK42	3	355	38.8	4.36	ACYL-COA binding family protein	ER
Glyma06G164700	6	GmANK43	2	739	83.58	7.61	ADP-ribosylation factor GTPase-activating protein	cyto
Glyma06G198600	6	GmANK44	2	156	17.43	6.32	Ankyrin repeat domain-containing protein	cyto
Glyma06G241300	6	GmANK45	5	418	46.98	6.57	Ankyrin repeat protein	cyto
Glyma06G242000	6	GmANK46	5	434	48.62	7.22	Ankyrin repeat protein	plas
Glyma06G294100	6	GmANK47	9	599	66.44	7.46	Ankyrin repeats-containing protein	plas
Glyma06G294200	6	GmANK48	9	665	74.99	8.64	Ankyrin repeats-containing protein	plas
Glyma06G294400	6	GmANK49	9	668	75.05	8.95	Ankyrin repeats-containing protein	plas
Glyma06G320200	6	GmANK50	3	353	37.88	4.41	Ankyrin-repeat protein	nucl
Glyma07G085800	7	GmANK51	3	481	53.87	6.3	Ankyrin repeat domain-containing protein	cyto
Glyma07G169500	7	GmANK52	6	526	57.73	8.41	Ankyrin repeat-containing protein	plas
Glyma07G185000	7	GmANK53	6	541	59.59	7.24	S-acyltransferase	plas
Glyma07G228000	7	GmANK54	2	422	47.78	7.95	Mitogen-activated protein kinase kinase kinase Raf17.1	cyto
Glyma07G234700	7	GmANK55	3	232	26.08	6.34	NA	cyto
Glyma07G242000	7	GmANK56	2	922	104.29	7.28	Calmodulin-binding transcription activator	nucl
Glyma07G261200	7	GmANK57	2	569	64.11	9.49	Ankyrin repeat family protein	chlo
Glyma08G008000	8	GmANK58	1	690	77.59	8.18	Signal recognition particle subunit SRP68	nucl
Glyma08G064300	8	GmANK59	6	542	59.68	7.26	S-acyltransferase	cyto
Glyma08G072100	8	GmANK60	2	1080	121.26	5.89	Calmodulin-binding transcription activator	nucl
Glyma08G079800	8	GmANK61	7	618	68.39	6.65	NA	plas
Glyma08G101800	8	GmANK62	9	677	75.04	8.35	Serine/threonine protein phosphatase 6 regulatory ankyrin repeat subunit C	nucl
Glyma08G105200	8	GmANK63	2	966	107.77	6	Calmodulin-binding family protein	nucl
Glyma08G126000	8	GmANK64	2	476	53.71	8.98	Ankyrin-kinase	mito
Glyma08G135200	8	GmANK65	2	1103	124.43	5.81	Ethylene-responsive calmodulin-binding family protein	nucl
Glyma08G150200	8	GmANK66	3	158	16.84	4.41	Ankyrin repeat domain-containing protein	cyto
Glyma08G178900	8	GmANK67	3	1082	121.61	5.72	Calmodulin-binding transcription activator	nucl
Glyma08G187600	8	GmANK68	4	835	95.46	7.2	Potassium channel	cyto
Glyma08G257600	8	GmANK69	2	824	92.1	7.32	ADP-ribosylation factor GTPase-activating protein AGD3	nucl
Glyma08G315500	8	GmANK70	6	347	38.05	8.24	Hop-interacting protein TH1015	cyto
Glyma08G358300	8	GmANK71	6	439	47.54	7.05	Hop-interacting protein TH1012	cyto
Glyma09G038300	9	GmANK72	1	912	102.97	6.84	Calmodulin-binding transcription activator	nucl
Glyma09G102900	9	GmANK73	3	729	78.39	6.12	Zinc finger CCCH domain-containing protein	nucl
Glyma09G143600	9	GmANK74	5	512	54.91	6.98	Ankyrin repeat-containing protein	nucl

(continued on next page)

Table 1 (continued)

Gene locus	Chr	Gene name	No. ANK	Length (aa)	MW (KD)	pI	BLAST2GO results	SL
Glyma09G191100	9	GmANK75	3	485	54.32	7.2	Ankyrin repeat domain-containing protein	cyto
Glyma09G214000	9	GmANK76	4	250	26.54	6.1	Ankyrin repeat domain-containing protein	nucl
Glyma09G215400	9	GmANK77	3	173	19.32	7.99	Ankyrin repeat domain-containing protein	nucl
Glyma10G059800	10	GmANK78	3	205	21.81	4.25	Ankyrin repeat family protein	cyto
Glyma10G097200	10	GmANK79	2	233	26.19	6.25	NA	cyto
Glyma10G100400	10	GmANK80	3	711	77.37	5.88	Zinc finger CCCH domain-containing protein	nucl
Glyma10G124500	10	GmANK81	3	558	62.93	9.38	Ankyrin repeat family protein	cysk
Glyma10G181100	10	GmANK82	2	457	48.88	6.68	Zinc finger family protein	cyto
Glyma10G238000	10	GmANK83	5	513	54.97	6.66	Ankyrin repeat-containing protein	nucl
Glyma10G291900	10	GmANK84	7	593	64.11	8.41	NA	plas
Glyma11G081800	11	GmANK85	5	445	50.57	7.24	Ankyrin repeat domain-containing protein	chlo
Glyma11G081900	11	GmANK86	4	422	47.83	9.84	Ankyrin repeat domain-containing protein	cyto
Glyma11G097200	11	GmANK87	3	383	41.8	4.82	Chaos family protein	chlo
Glyma11G104500	11	GmANK88	2	777	87.81	6.28	Arf GTPase-activating domain-containing protein	nucl
Glyma11G140000	11	GmANK89	6	448	48.31	7.35	Ankyrin repeat family protein	nucl
Glyma11G157400	11	GmANK90	9	1638	179.85	5.95	E3 ubiquitin-protein ligase KEG	chlo
Glyma11G174400	11	GmANK91	2	606	65.79	6.44	Zinc finger CCCH domain-containing protein	nucl
Glyma11G247700	11	GmANK92	8	653	71.9	7.25	Serine/threonine protein phosphatase 6 regulatory ankyrin repeat subunit C	chlo
Glyma11G248800	11	GmANK93	3	359	39.27	9.03	Ankyrin repeat domain-containing protein	chlo
Glyma11G251900	11	GmANK94	2	911	10.24	5.62	Calmodulin-binding transcription activator	nucl
Glyma12G029500	12	GmANK95	2	777	87.63	6.28	Arf GTPase-activating domain-containing protein	nucl
Glyma12G063600	12	GmANK96	6	448	48.44	7.49	Ankyrin repeat family protein	nucl
Glyma12G074900	12	GmANK97	8	549	59.13	9.48	Ankyrin repeat-containing protein	plas
Glyma12G111400	12	GmANK98	9	658	73.75	9.12	Ankyrin repeats-containing protein	plas
Glyma12G170500	12	GmANK99	4	832	95.07	7.68	Potassium channel	cyto
Glyma12G237300	12	GmANK100	3	705	75.82	6.49	Zinc finger CCCH domain-containing protein	nucl
Glyma13G091900	13	GmANK101	5	509	54.97	7.28	Ankyrin repeat-rich protein	chlo
Glyma13G130300	13	GmANK102	6	470	52.46	8.89	Ankyrin repeat family protein	plas
Glyma13G146500	13	GmANK103	3	205	21.92	4.26	Ankyrin repeat family protein	mito
Glyma13G152400	13	GmANK104	5	331	36.96	4.87	Ankyrin repeat family protein	chlo
Glyma13G163400	13	GmANK105	5	633	69.06	6.71	S-acyltransferase	plas
Glyma13G195200	13	GmANK106	9	1422	157.54	5.72	E3 ubiquitin-protein ligase KEG	cyto
Glyma13G202000	13	GmANK107	5	332	37.23	8.42	Ankyrin repeat family protein	cyto
Glyma13G202800	13	GmANK108	3	702	75.66	6.75	Zinc finger CCCH domain-containing protein	nucl
Glyma13G223400	13	GmANK109	3	456	51.45	7.44	Ankyrin protein kinase	cyto
Glyma13G331500	13	GmANK110	8	546	59.38	9.04	Ankyrin repeat-containing protein	plas
Glyma13G335300	13	GmANK111	7	452	48.55	7.35	Ankyrin repeat family protein	nucl
Glyma14G016300	14	GmANK112	2	681	74.6	6.88	Zinc finger CCCH domain-containing protein	nucl
Glyma14G026700	14	GmANK113	3	454	51.34	7.49	Protein kinase family protein	mito
Glyma14G099600	14	GmANK114	2	318	34.92	10.12	Calmodulin-binding transcription factor SR2L	nucl
Glyma14G126500	14	GmANK115	6	876	98.94	6.61	Shaker-like potassium channel 1	plas
Glyma14G194100	14	GmANK116	2	691	75.13	6.01	Zinc finger CCCH domain-containing protein	nucl
Glyma14G212500	14	GmANK117	5	851	97.56	6.97	Potassium channel SKOR	plas
Glyma15G018600	15	GmANK118	8	648	72.03	8.4	Ankyrin repeat-containing protein	plas
Glyma15G039100	15	GmANK119	7	445	47.87	7.05	Ankyrin repeat family protein	nucl
Glyma15G042600	15	GmANK120	8	546	59.22	8.85	Ankyrin repeat-containing protein	mito
Glyma15G053600	15	GmANK121	3	1089	122.48	5.87	Calmodulin-binding transcription activator	nucl
Glyma15G088500	15	GmANK122	3	457	51.4	7.92	Ankyrin protein kinase	pero
Glyma15G143400	15	GmANK123	1	912	103.08	7.68	Calmodulin-binding transcription activator	nucl
Glyma15G233000	15	GmANK124	2	234	25.96	8.87	A subunit of NADH dehydrogenase	cyto
Glyma15G274800	15	GmANK125	2	723	77.74	5.95	Zinc finger CCCH domain-containing protein	nucl
Glyma16G038200	16	GmANK126	7	517	56.21	7.29	NA	cyto
Glyma16G038900	16	GmANK127	2	1081	117.59	9.09	Ankyrin repeat domain-containing protein	nucl
Glyma16G060400	16	GmANK128	11	754	81.32	6.58	Ankyrin repeat domain-containing protein	cyto
Glyma16G062700	16	GmANK129	8	669	73.63	9.27	Ankyrin-like protein	nucl
Glyma16G196400	16	GmANK130	5	512	54.88	6.82	Ankyrin repeat-containing protein	nucl
Glyma17G012800	17	GmANK131	2	563	63.25	9.21	Ankyrin repeat family protein	chlo
Glyma17G031900	17	GmANK132	2	923	103.87	7.51	Calmodulin-binding transcription activator	nucl
Glyma17G068400	17	GmANK133	5	511	55.3	7.53	Ankyrin repeat-rich protein	chlo
Glyma17G107500	17	GmANK134	5	634	69.58	6.73	S-acyltransferase	plas
Glyma17G118300	17	GmANK135	5	876	98.11	7.3	Potassium channel	plas
Glyma17G207900	17	GmANK136	5	867	97.56	6.23	Shaker-like potassium channel 1	plas
Glyma18G005100	18	GmANK137	2	963	108.04	5.4	Calmodulin-binding family protein	nucl
Glyma18G008500	18	GmANK138	11	762	83.11	7.5	Ankyrin repeat domain-containing protein	cyto
Glyma18G009500	18	GmANK139	8	652	71.55	6.87	Serine/threonine protein phosphatase 6 regulatory ankyrin repeat subunit C	nucl
Glyma18G044300	18	GmANK140	5	332	36.81	5.11	Ankyrin repeat family protein	chlo
Glyma18G062200	18	GmANK141	2	609	66.47	5.96	Zinc finger CCCH domain-containing protein	nucl
Glyma18G100300	18	GmANK142	5	727	81.37	8.86	Nuclear factor NF-kappa-B p105 subunit	plas
Glyma18G176600	18	GmANK143	6	442	47.78	6.96	Hop-interacting protein THIO12	nucl
Glyma18G282000	18	GmANK144	2	840	93.81	7	ADP-ribosylation factor GTPase-activating protein AGD3	nucl
Glyma19G086700	19	GmANK145	11	755	81.24	6.51	Ankyrin repeat domain-containing protein	cyto
Glyma19G113700	19	GmANK146	2	1078	117.28	8.99	Ankyrin repeat domain-containing protein	nucl
Glyma19G114600	19	GmANK147	8	528	57.52	7.53	NA	chlo
Glyma19G131000	19	GmANK148	2	414	45.48	6.54	BTB/POZ ankyrin repeat protein	cyto
Glyma19G141700	19	GmANK149	2	668	73.47	6.23	Zinc finger CCCH domain-containing protein	nucl
Glyma19G171500	19	GmANK150	5	311	34.7	8.28	Ankyrin repeat family protein	nucl

Table 1 (continued)

Gene locus	Chr	Gene name	No. ANK	Length (aa)	MW (KD)	pI	BLAST2GO results	SL
Glyma19G171600	19	GmANK151	3	270	30.61	7.84	Ankyrin repeat family protein	nucl
Glyma19G175300	19	GmANK152	7	582	63.94	9.55	Ankyrin repeat-containing protein	plas
Glyma19G175400	19	GmANK153	8	531	58.2	9.08	Ankyrin repeat-containing protein	plas
Glyma19G245700	19	GmANK154	7	461	49.04	5.04	Ankyrin repeat family protein	chlo
Glyma19G263700	19	GmANK155	8	559	61.46	9.51	Ankyrin repeats-containing protein	plas
Glyma20G031300	20	GmANK156	2	424	48	7.68	Mitogen-activated protein kinase kinase Raf17.1	cyto
Glyma20G076600	20	GmANK157	2	549	61.65	9.44	Ankyrin repeat family protein	nucl
Glyma20G156600	20	GmANK158	5	513	54.93	6.87	Ankyrin repeat-containing protein	chlo
Glyma20G209400	20	GmANK159	2	458	49.03	5.96	Zinc finger family protein	cyto
Glyma20G241200	20	GmANK160	7	586	63.58	8.83	NA	plas
GlymaU025300	X	GmANK161	6	453	50.14	8.9	Ankyrin repeat protein	ER
GlymaU033200	X	GmANK162	8	550	59.2	9.42	Ankyrin repeat-containing protein	plas

Note: Chromosome: Chr, number of ANK: No. ANK, length (amino acids): Length (aa), molecular weight (kiloDalton): MW (KD), isoelectric points: pI, X presents the unknown, SL: sub-cellular localization.

2.8. Statistical analysis

Data were analyzed using an ANOVA test in Microsoft Excel. Significant differences among means were determined by * at $p < 0.05$ and ** at $p < 0.01$.

3. Results

3.1. Genome-wide identification of proteins containing ANK domain in soybean genome

The HMM profile (PF00023) was used to search for conserved ANK domain in soybean genome data, totally, 162 putative ANK-containing proteins were identified and validated by on-line SMART software and pfam and BLAST2GO analysis. The gene locus and Chromosome (Chr) localization were provided and gene names (GmANK1–162) were designated according to the order of Chr1–20, indicating that the GmANKs distributed in each Chr, and the Chr8 (14) contained the most, while he Chr20 included 5 GmANKs. Furthermore, the peptide length, number

of ANK domain, molecular weight (MW) and isoelectric points (pI) were deduced from the protein sequences respectively. In details, the largest gene (*Glyma11G157400*, *GmANK90*) encoded 1638 amino acids with the MW of 179.85, while the shortest one (*Glyma02G033700*, *GmANK6*) contained only 143 with a MW of 15.86. The number of ANK domain ranged from 1–11 and the average is 4.5. The protein sub-cellular localization of GmANKs mainly distributed nucl (nucleus, 35.19%), cyto (cytoplasm, 22.22%), plas (plasma membrane, 22.22%) and chlo (chloroplast, 12.35%), the remaining localized other organelles, such as mitochondrion, ER (endoplasmic reticulum) (Table 1).

Based on the results of SMART analysis for every GmANKs, we found that ANK domain mainly combined with the BAR (Endocytosis and intracellular transport), PH (Pleckstrin homology domain), ArfGap (Putative GTP-ase activating proteins for the small GTPase), BTB (Broad-Complex, Tramtrack and Bric a brac), ZnFC3H1 (Zinc fingerC3H1), TM (Transmembrane region), cNMP (Cyclic nucleotide-monophosphate binding domain), TPR (Tetratricopeptide repeats), RING (Ring finger), CG1 (CALModulin-binding Transcription Activator), IQ (Calmodulin-binding motif), AAA (ATPases associated with a variety of cellular

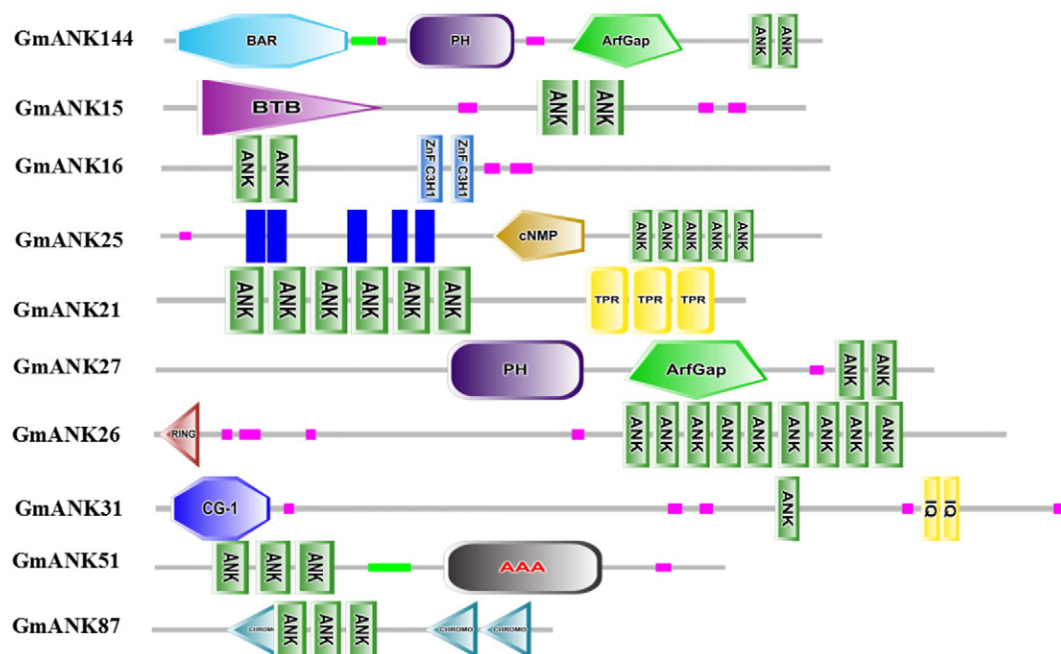


Fig. 1. Structure and domain composition of representative GmANK proteins from each subfamily. Domain abbreviations are: ANK ankyrin repeat domain; TM transmembrane; TPR tetratricopeptide repeat domain; RING ring finger domain; ZnF zinc finger; BTB broad-complex, tramtrack and bric a brac domains; IQ calmodulin-binding domain; BAR BAR domain; PH pleckstrin homology domain; ArfGap putative GTP-ase activating proteins for the small GTPase, ARF; AAA ATPase family associated with various cellular activities; CHROMO CHROMatin organization MOdifier domain; cNMP cyclic nucleotide-binding domain; CG1 CALModulin-binding transcription activator. The length and order of domains represent actual situation in each protein.

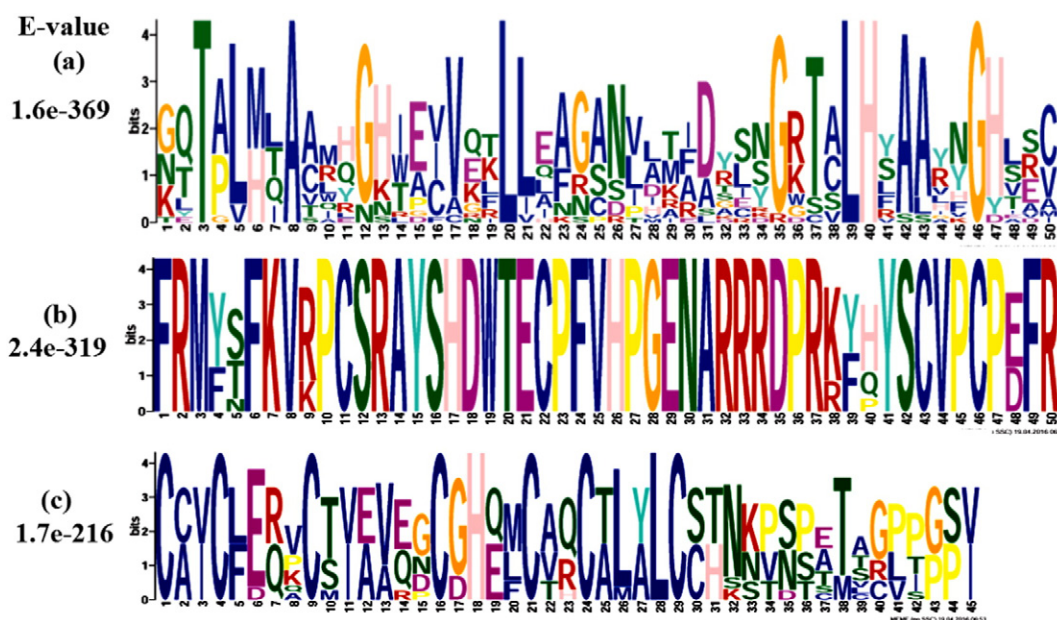


Fig. 2. Sequence logos for the GmANK repeats motifs of soybean. The number of X-axis represent the sequence positions in the ANK repeats motifs. The Y-axis represents the information content measured in bits.

activities) and CHROMO (CHRomatin Organization Modifier) domains, indicating the functional diversity of ANK (Fig. 1).

To reveal the sequence conservation of soybean ANK repeats motifs, we performed MEME analysis of GmANK1–80 to generate sequence logos, which showed that there were several highly conserved amino acid residues within the ANK repeats motifs, including the T at position 3, the A at position 8, the L at position 20, LH at position 39–40 and G at position 46 respectively (Fig. 2). Each logo of the ANK motifs was similar to the Pfam sequence (PF00023: DGFTPLHLAALRGNEVVK LLSQGADLNAQDD).

3.2. In silico expression patterns analysis of GmANKs genes in soybean different tissues

To insight into the expression patterns of GmANKs genes in soybean different tissues, the expression data in different tissues was downloaded from soybase (<http://soybase.org>) and analyzed in silico and displayed using Java-Treeview software. The heatmap showed that some GmANKs had no expression in any tissues, such as GmANK18,27,45,46,61,107,118,150,151, some GmANKs were organ-specific, including GmANK9,17 in nodule, GmANK19,152 in flower and GmANK48,49,70,102,142 in root respectively. Some GmANKs were constitutively expressed, including GmANK25,90,100,108. Seven GmANKs(32,44,79,93,124,161,162) had no expression data in soybean database (Fig. 3).

3.3. Phylogenetic analysis of GmANKs

To gain the phylogenetic relationship among the GmANKs proteins and acquire the evolutionary history of this gene family, a phylogenetic tree was constructed except for the GmANK2,44,55,79 with the low similarities, 158 GmANKs were clustered using MEGA5.0 software. The results indicated that GmANK proteins were divided into seven distinct major groups (I–VII), and some major groups could be divided into different subgroups. In general, most of the closely related members in the phylogenetic tree had the same or very similar domain composition, such as Zinc finger CCCH domain-containing protein (Group VI) and Ankyrin-kinase family (Group VII) (Fig. 4).

3.4. Identification of GmANKs proteins related to abiotic stresses in soybean

Based on our digital gene expression profiling (DGEP) data (Ali et al., 2012, unpublished data), the GmANK genes which had the response to drought (PEG treatment) and salinity stresses were screened and identified both in *Glycine max* and *Glycine soja*. The results demonstrated that some GmANKs (GmANK47,154,103,101,66,102,4,25) showed the response to drought (Fig. 5,a), and some GmANKs(75,10,103,66,47,101,154,4,2) maybe played important roles in responding to salinity stress (Fig. 5,b).

3.5. The cloning and expression patterns of GmANK39-like(GmANK6)

GmANK6 was a shortest one in soybean ANK family and was isolated by screening a salt-treated yeast two hybrid library in our lab previously. BLASTx search showed that GmANK6 had a higher identities with GmANK-39-Like, and had a 75%, 68% and 47% identities with *Cajanus cajan* (CcAnk2 accession no:KYP71517), *Cicer arietinum* (CaAnk50, accession no:XP_004496053) and *Prunus mume* (ankyrin-3-like, accession no:XP_008234958) respectively. SMART on-line software analysis indicated that the GmANK-39-Like protein sequence only contains two Ankyrin repeats domains from amino acids 47–76 and 80–108. However, the others above mentioned possessed five Ankyrin repeats domains and had a relatively conserved sequence characterization at Ankyrin repeats domain sites for these proteins (Supplemental Fig. 1).

Phylogenetic analysis was used to find out the relationship among the GmANK-39-Like and other ANKs proteins from *Arabidopsis thaliana* (AtAnk2), *Cicer arietinum* (CaAnk50), *Cajanus cajan* (CcAnk2), *Gossypium raimondii* (GrAnk50), *Medicago truncatula* (MtAnk), *Ricinus communis* (RcAnk1), *Theobroma cacao* (TcAnk), *Vigna radiata* (VrAnk1) and *Vitis vinifera* (VvAnk65). GmANK-39-Like was found to cluster together with other ANKs and had a higher homology with VrAnk1 and CcAnk2, however, had the lower homology with AtAnk2, indicating that GmANK-39-Like really originates from the legume (Supplemental Fig. 2).

To examine the tissue expression pattern of GmANK-39-Like, the root, stem, leaf, flower and pod were investigated. The result showed that GmANK-39-Like mainly expressed the root, leaf and pod, weak in stem and flower (Fig. 6 a), which was consistent with the data of in silico

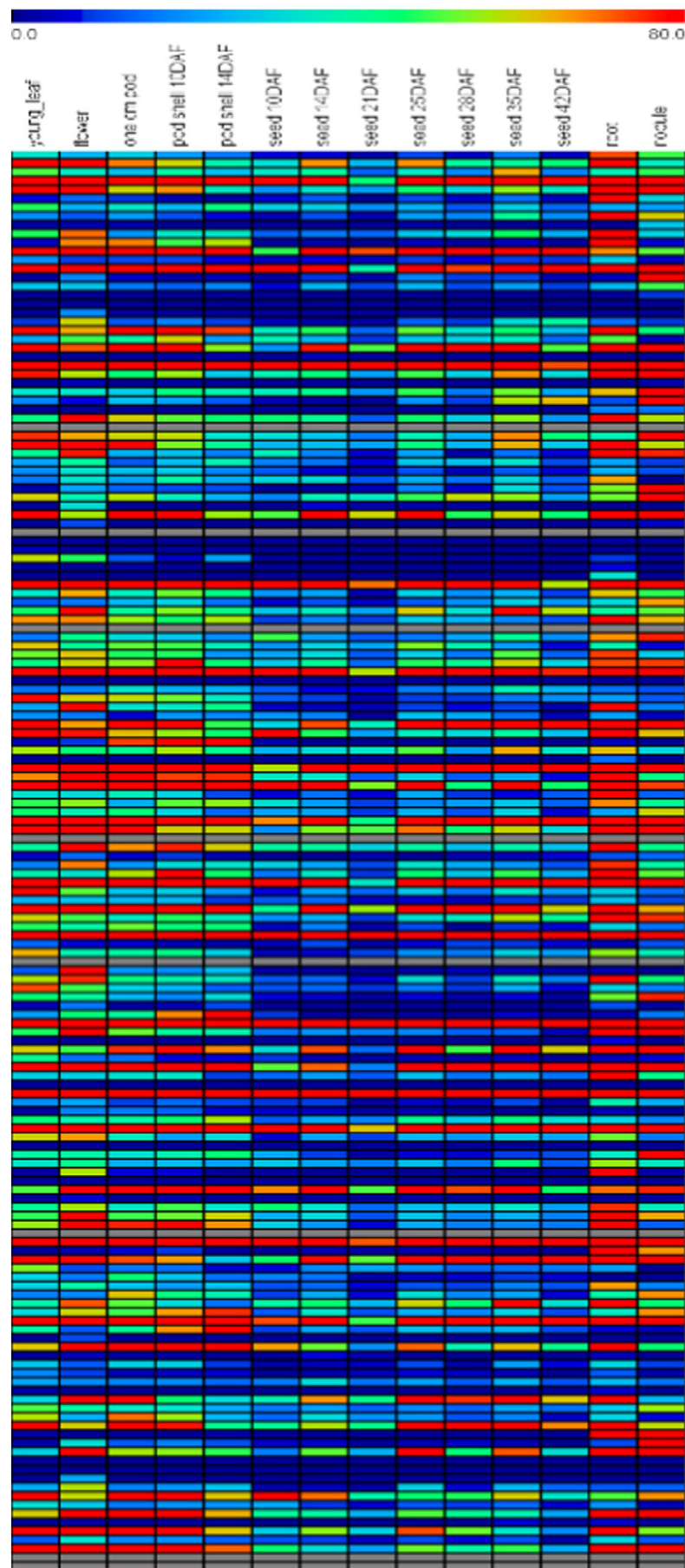


Fig. 3. Phylogenetic relationships of 158 soybean ANK repeats proteins. The tree was constructed using the MEGA5.0 software.

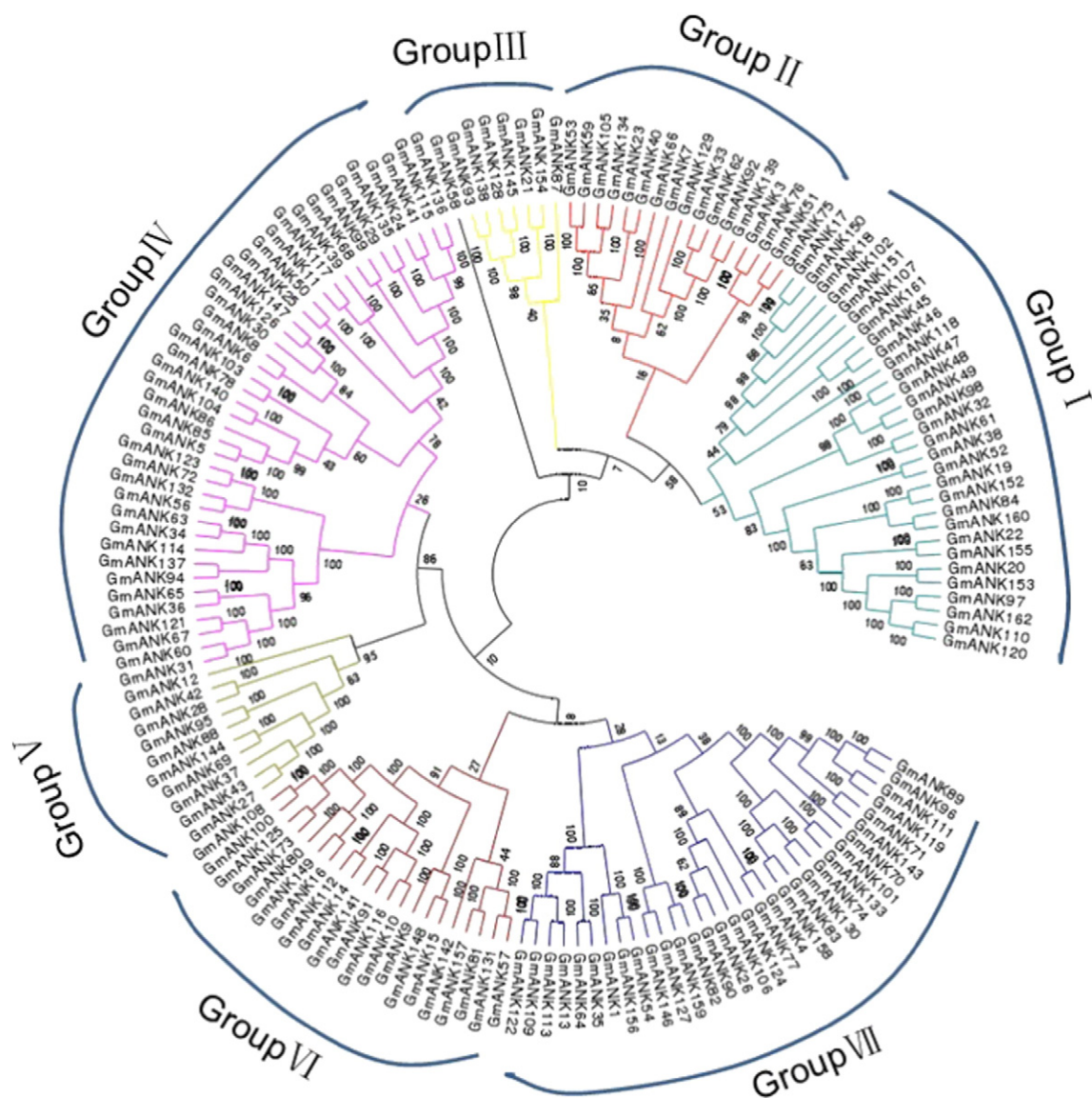


Fig. 4. Expression analysis of *GmANK* genes in the different developmental stages of soybean. Hierarchical cluster displaying expression profiles for 162 *GmANK* genes in 14 tissues or developmental stages. The tissues and color are indicated on the top of expression heatmap. Different color lines representing different expression patterns for each gene.

expression. To determine the expression pattern of *GmANK-39-Like* under abiotic stresses, the soybean seedlings of three-leaf stage were treated with 20% PEG and 200 mM NaCl solutions for different time points, then QRT-PCR was carried out using treated roots tissue as template. The results showed that the expression of *GmANK-39-Like* was increased at 2 h, then significantly declined at 4 h and went back to previous expression level at 8 h (Fig. 6 b). However, the expression under salinity stress was decreased all the time points (Fig. 6 c).

3.6. Subcellular localization of *GmANK39-like* (*GmANK6*)

To examine the functional localization of *GmANK-39-Like*, the PEG-mediated protoplast transformation method was performed. Enhanced green fluorescent protein (EGFP) in-frame fused C terminally to *GmANK-39-Like* and only EGFP (empty vector) were transiently expressed in *Arabidopsis* mesophyll protoplast driven by cauliflower mosaic virus 35S promoter respectively. The results of subcellular localization demonstrated that GFP was detected all the whole cells for the 35S::GFP, but mainly distributed cytoplasm for 35S::*GmANK-39-Like*.

Like:GFP (Fig. 7), indicating *GmANK-39-Like* is not a membrane protein and functions at the cytoplasm. (See Fig. 7.)

4. Discussion

Soil salinity becomes more and more one of the major obstacles in the world. In China, >600 ha soil field were increased every year in JiangSu province because of sea movement, and in Australia, large amounts of salt accumulated from cyclical deposition via rainfall, weathering of saline material and connate salts laid down as marine sediments in earlier geological time (Spies and Woodgate, 2004). The activities of man, including agronomy and hydrology, can also cause soil salinization via mobilisation of salts (Huang et al., 2016). Plant directly touched with the saline soil, therefore, understanding the principles of plant-soil interaction and revealing the mechanisms of plant response to salt stress will provide the important theoretical basis for modern salt-soil agriculture.

The ankyrin repeats are about 33-amino-acid in length repeats found in numerous proteins, and serve as domains for protein-protein interactions (Michaely and Bennett, 1992). The ankyrin repeat was

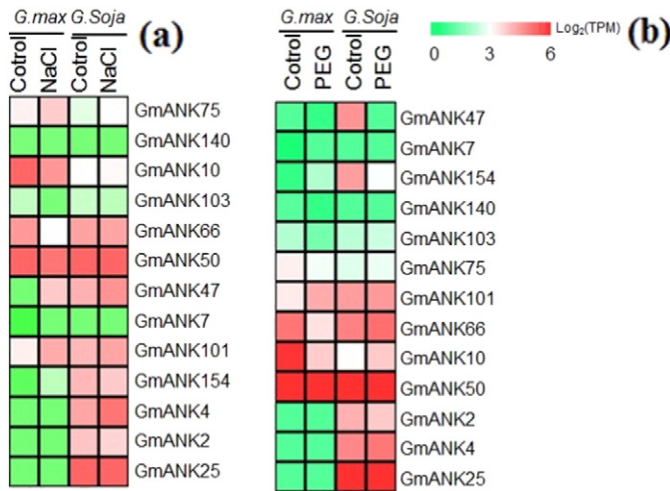


Fig. 5. Normalized expression of differentially expressed *GmANK* genes in *Glycine soja* and *Glycine max*. (a) Heatmap of differentially expressed *GmANK* genes based on their relative gene expression values (Z-score) computed using DGEP data under 0 and 200 mM NaCl stress. (b) Heatmap of differentially expressed *GmANK* genes based on their relative gene expression values (Z-score) computed using DGEP data under 0 and 20% PEG stress. Red to green color indicates high to low expression levels.

first identified in the yeast cell cycle regulator Swi6/Cdc10 and the *Drosophila* signaling protein Notch (Breedon and Nasmyth, 1987). The number of ankyrin repeats varies for different proteins. Analysis of the SMART and PFAM databases show that the number of repeats per protein ranges from 1 to 33. To date, the protein with the most ankyrin repeats is the ORF EAA39756 from *Giardia lamblia* containing 34 repeats when PFAM and SMART predictions are combined (Mosavi et al.,

2004). In the present study, we found the most ankyrin repeats in soybean is Glyma16G060400, Glyma18G008500 and Glyma19G086700, which contain 11 repeats. However, the least ones (Glyma01G052600, Glyma05G117000, Glyma08G008000, Glyma09G038300, Glyma15G143400) only include one repeat. The average of the ankyrin repeats is 4.5, which is identical with the rice ANK proteins (Becerra et al., 2004). These may indicate that the structure of ankyrin repeats is of importance for ANK proteins, especially for protein-protein interactions.

In recent years, the number of ANK proteins has been found to be 175 in rice, 105 in the *Arabidopsis thaliana*, 130 in tomato and 71 in maize (Becerra et al., 2004; Huang et al., 2009; Yuan et al., 2013; Jiang et al., 2013). In this study, we identified 162 ANK proteins in soybean, which can be classified into 11 subfamilies according to domain composition and they are distributed on all 20 chromosomes. Moreover, some ANK protein showed gene duplications on some chromosomes, such as Glyma06G294100, Glyma06G294200 and Glyma06G294400. This is the first report about the ANK family in soybean.

In addition, ANK domain mainly combined with the other domains (Fig. 1). Huang et al. (2009) found that rice ANK proteins contained Motile_Sperm MSP (major sperm protein) domain, ACBP (acyl CoA binding Protein), PPR (pentatricopeptide repeats); Ion_trans (ion transport Protein), RCC1 (regulator of chromosome condensation); rve (integrase core domain); RVT (reverse transcriptase RNA-dependent DNA polymerase); RRM (RNA recognition motif); ubiquitin (ubiquitin family) instead of domains mentioned in this study. This inferred that rice ANKs possessed more diverse functions than maize and soybean, which was consistent with the statement that ANK repeats mediate protein-protein interactions also in plants (Becerra et al., 2004).

In plants, one of the first characterized ANK proteins was AKR (*Arabidopsis* Ankyrin Repeat). The transcripts levels of AKR were found to be light dependent and was associated with the regulation of chloroplast differentiation (Zhang et al., 1992). To insight into the

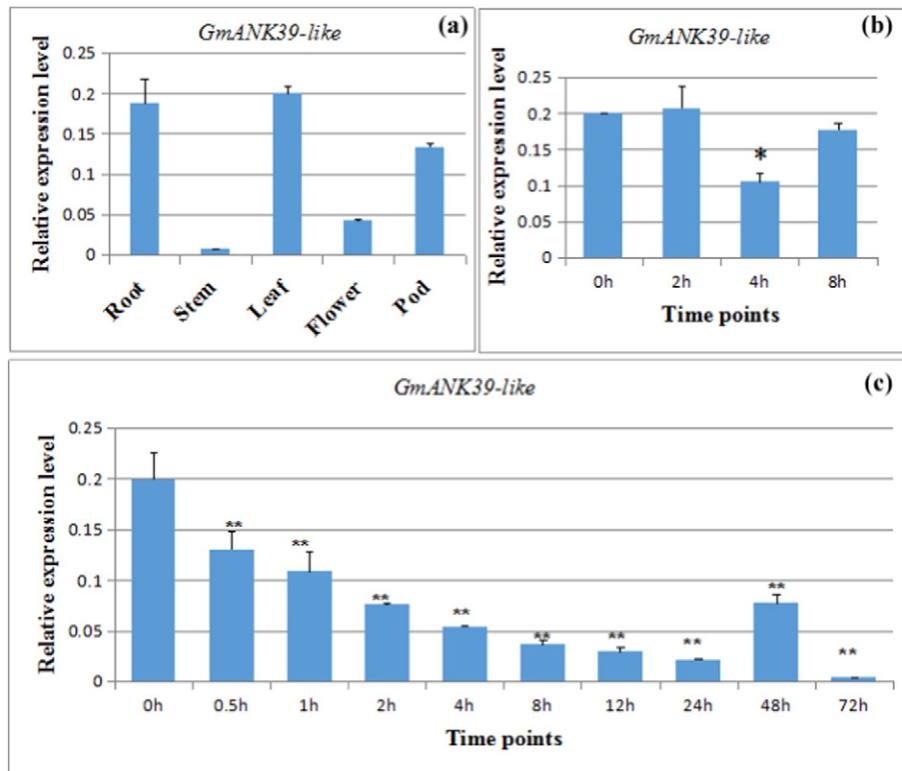


Fig. 6. The expression pattern of *GmANK39-Like* in different tissues and under different abiotic stresses in soybean. (a) The expression pattern of *GmANK39-Like* in root, stem, leaf, flower and pod at seedling, flowering and podding stages, respectively using quantitative RT-PCR. (b) The expression pattern of *GmANK39-Like* in soybean roots under PEG6000 for 0 h, 2 h, 4 h and 8 h. (c) The expression pattern of *GmANK39-Like* in soybean roots under 200 mM NaCl for 0 h, 1 h, 2 h, 4 h, 8 h, 12 h, 24 h, 48 h and 72 h. *p < 0.05 level; **p < 0.01.

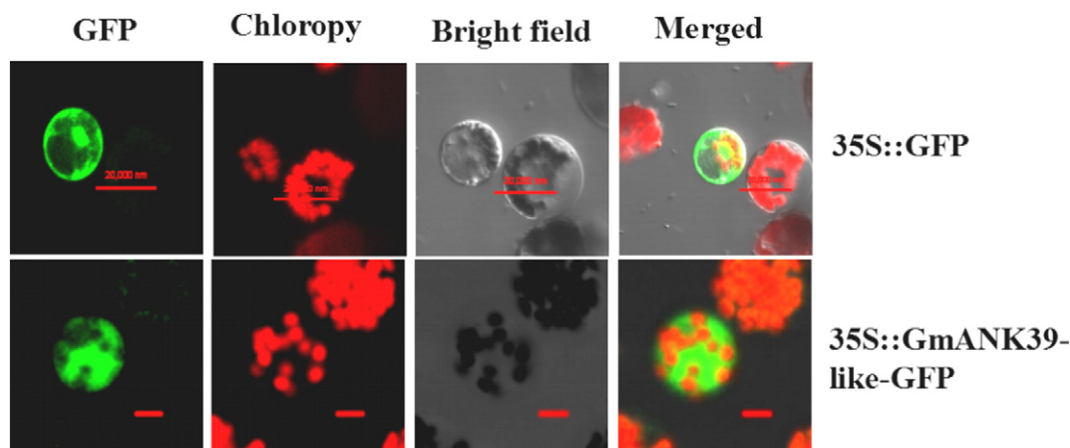


Fig. 7. Subcellular localization of GmANK-39-Like protein in *Arabidopsis* protoplasts. The GmANK-39-Like-green fluorescent protein (p35S::GmANK-39-Like-GFP) fusion construct was generated by inserting GmANK-39-Like into pJIT166-GFP without a termination codon to creating an in-frame fusion between the CDS and GFP. p35S::GmANK-39-Like-GFP and the GFP control plasmid (p35S::GFP) were transformed into *Arabidopsis* protoplasts by the PEG4000-mediated method. Subcellular localization demonstrated that GFP was detected all the whole cells for the 35S::GFP, but mainly distributed cytoplasm for 35S::GmANK-39-Like:GFP. Scale bars = 20 μ m.

expression profile of *GmANK* genes which will provide clues to the function of each gene, the in silico expression patterns for each gene were performed. The data showed that the diverse or different expression types, including organ-specific, constitutive and developmental stage-specific, indicating that they play important roles in plant development and response to stresses. In this study, the GmANK6 showed an increase expression after drought treatment at early stage, however, a decreasing after salt treatment, indicating ANK proteins have the response to abiotic stresses. The cytoplasm localization of GmANK6 complied with the putative result by WoLF PSORT tool. The results will provide important clues to explore ANK genes expression and function in future studies in soybean.

5. Conclusions

To mining the key salt tolerance related genes in soybean for planting in salt soil, 162 *GmANK* genes were identified by database searches and classified by protein domains, the gene locus and Chromosome (Chr) localization the protein length, number of ANK domain, molecular weight, isoelectric points, expression patterns and phylogenetic relationship of the ANK genes in soybean were analyzed, and the expression patterns of *GmANK39-like*(*GmANK6*) gene in the developmental stages of soybean and its responses to abiotic stress and subcellular localization were also surveyed.

Author contributions

DY. Zhang, and Q. Wan conceived, designed and conducted the experiments. ZL. Xu, XL. He, and HY. Huang analyzed the data and results. DY. Zhang, wrote the manuscript. HB. Shao and DY. Zhang monitored the experiments and critically commented on the manuscript. All authors read and approved the final manuscript.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgments

This study was sponsored by the National Science Foundation of China (31101166); the Jiangsu Natural Science Foundation, China (BK20151364), Shuangchuang Talent Plan of Jiangsu Province, China and Jiangsu Agriculture Science and Technology Innovation Fund (JASTIF, CX(15)1005).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.scitotenv.2016.06.078>.

References

- Abel, S., Theologis, A., 1994. Transient transformation of *Arabidopsis* leaf protoplasts: a versatile experimental system to study gene expression. *Plant J.* 5, 421–427. <http://dx.doi.org/10.1111/j.1365-313x.1994.00421.x>.
- Ali, Z., Zhang, D.Y., Xu, Z.L., Xu, L., Yi, J.X., He, X.L., Huang, Y.H., Liu, X.Q., Khan, A.A., Trethowan, R.M., Ma, H.X., 2012. Uncovering the salt response of soybean by unraveling its wild and cultivated functional genomes using tag sequencing. *PLoS one* 7, e48819. <http://dx.doi.org/10.1371/journal.pone.0048819>.
- Bae, W., Lee, Y.J., Kim, D.H., Lee, J., Kim, S., Sohn, E.J., Hwang, I., 2008. AKR2A-mediated import of chloroplast outer membrane proteins is essential for chloroplast biogenesis. *Nat. Cell Biol.* 10, 220–227. <http://dx.doi.org/10.1038/ncb1683>.
- Bailey, T.L., Williams, N., Misleh, C., Li, W.W., 2006. MEME: discovering and analyzing DNA and protein sequence motifs. *Nucleic Acids Res.* 34, 369–373. <http://dx.doi.org/10.1093/nar/gkl198>.
- Becerra, C., Jahrmann, T., Puigdomenech, P., Vicient, C.M., 2004. Ankyrin repeat-containing proteins in *Arabidopsis*: characterization of a novel and abundant group of genes coding ankyrin-transmembrane proteins. *Gene* 340, 111–121. <http://dx.doi.org/10.1016/j.gene.2004.06.006>.
- Breeden, L., Nasmyth, K., 1987. Similarity between cell-cycle genes of budding yeast and fission yeast and the Notch gene of *Drosophila*. *Nature* 329, 651–654. <http://dx.doi.org/10.1038/329651a0>.
- Ha, C.M., Jun, J.H., Nam, H.G., Fletcher, J.C., 2004. BLADE-ON-PETIOLE1 encodes a BTB/POZ domain protein required for leaf morphogenesis in *Arabidopsis thaliana*. *Plant Cell Physiol.* 45, 1361–1370. <http://dx.doi.org/10.1093/pcp/pch201>.
- Huang, J., Prochazka, M.J., Triantafyllis, J., 2016. Irrigation salinity hazard assessment and risk mapping in the lower Macintyre Valley, Australia. *Sci. Total Environ.* 551–552, 460–473. <http://dx.doi.org/10.1016/j.scitotenv.2016.01.200>.
- Huang, J., Zhao, X., Yu, H., Ouyang, Y., Wang, L., Zhang, Q., 2009. The ankyrin repeat gene family in rice: genome-wide identification, classification and expression profiling. *Plant Mol. Biol.* 71, 207–226. <http://dx.doi.org/10.1007/s11103-009-9518-6>.
- Ito, K., Ishii, N., Miyashita, A., Tominaga, K., Kuriyama, H., Maruyama, H., Shirai, M., Naito, M., Arakawa, M., Kuwano, R., 1999. Molecular cloning of a novel 130-kDa cytoplasmic protein, Ankhzn, containing ankyrin repeats hooked to a zinc finger motif. *Biochem. Biophys. Res. Commun.* 257, 206–213. <http://dx.doi.org/10.1006/bbrc.1999.0430>.
- Jiang, H., Wu, Q., Jin, J., Sheng, L., Yan, H., Cheng, B., Zhu, S., 2013. Genome-wide identification and expression profiling of ankyrin-repeat gene family in maize. *Dev. Genes. Evol.* 223, 303–318. <http://dx.doi.org/10.1007/s00427-013-0447-7>.
- Lu, H., Liu, Y., Greenberg, J.T., 2005. Structure-function analysis of the plasma membrane-localized *Arabidopsis* defense component ACD6. *Plant J.* 44, 798–809. <http://dx.doi.org/10.1111/j.1365-313x.2005.02567.x>.
- Michaely, P., Bennett, V., 1992. The ANK repeat: a ubiquitous motif involved in macromolecular recognition. *Trends Cell Biol.* 2, 127–129. [http://dx.doi.org/10.1016/0962-8924\(92\)90084-Z](http://dx.doi.org/10.1016/0962-8924(92)90084-Z).
- Mosavi, L.K., Cammett, T.J., Desrosiers, D.C., Peng, Z.Y., 2004. The ankyrin repeat as molecular architecture for protein recognition. *Protein Sci.* 13, 1435–1448. <http://dx.doi.org/10.1110/ps.03554604>.
- Sakamoto, H., Matsuda, O., Iba, K., 2008. ITN1, a novel gene encoding an ankyrin-repeat protein that affects the ABA-mediated production of reactive oxygen species and is

- involved in salt-stress tolerance in *Arabidopsis thaliana*. Plant J. 56, 411–422. <http://dx.doi.org/10.1111/j.1365-3113.2008.03614.x>.
- Sedgwick, S.G., Smerdon, S.J., 1999. The ankyrin repeat: a diversity of interactions on a common structural framework. Trends Biochem. Sci. 24, 311–316. [http://dx.doi.org/10.1016/S0968-0004\(99\)01426-7](http://dx.doi.org/10.1016/S0968-0004(99)01426-7).
- Seong, E.S., Choi, D., Cho, H.S., Lim, C.K., Cho, H.J., Wang, M.H., 2007. Characterization of a stress-responsive ankyrin repeat-containing zinc finger protein of *Capsicum annuum* (CaKR1). J. Biochem. Mol. Biol. 40, 952–958. <http://dx.doi.org/10.5483/BMBRep.2007.40.6.952>.
- Sharma, M., Pandey, G.K., 2015. Expansion and function of repeat domain proteins during stress and development in plants. Front. Plant Sci. 6, 1218. <http://dx.doi.org/10.3389/fpls.2015.01218>.
- Spies, B., Woodgate, P., 2004. Review of methods for mapping dry land salinity. Aseg. Extend. Abstr. 2004, 1–4.
- Tamura, K., Peterson, D., Peterson, N., Stecher, G., Nei, M., Kumar, S., 2011. MEGA5: molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. Mol. Biol. Evol. 28, 2731–2739. <http://dx.doi.org/10.1093/molbev/msr121>.
- Xiong, L., Lee, H., Ishitani, M., Zhu, J.K., 2002. Regulation of osmotic stress-responsive gene expression by the LOS6/ABA1 locus in *Arabidopsis*. J. Biol. Chem. 277, 8588–8596. <http://dx.doi.org/10.1074/jbc.M109275200>.
- Yan, J., Wang, J., Zhang, H., 2002. An ankyrin repeat-containing protein plays a role in both disease resistance and antioxidation metabolism. Plant J. 29, 193–202. <http://dx.doi.org/10.1046/j.0960-7412.2001.01205.x>.
- Yoo, S.D., Cho, Y.H., Sheen, J., 2007. *Arabidopsis* mesophyll protoplasts: a versatile cell system for transient gene expression analysis. Nat. Protoc. 2, 1565–1572. <http://dx.doi.org/10.1038/nprot.2007.199>.
- Yuan, X., Zhang, S., Qing, X., Sun, M., Liu, S., Su, H., Shu, H., Li, X., 2013. Superfamily of ankyrin repeat proteins in tomato. Gene 523, 126–136. <http://dx.doi.org/10.1016/j.gene.2013.03.122>.
- Zhang, D.Y., Ali, Z., Wang, C.B., Xu, L., Yi, J.X., Xu, Z.L., Liu, X.Q., He, X.L., Huang, Y.H., Khan, I.A., Ma, H.X., 2013. Genome-wide sequence characterization and expression analysis of major intrinsic proteins in soybean (*Glycine max* L.). PLoS ONE 8, e56312. <http://dx.doi.org/10.1371/annotation/e3307d0c-bb59-4f75-89b4-8e0d5af087d5>.
- Zhang, H., Scheirer, D.C., Fowle, W.H., Goodman, H.M., 1992. Expression of antisense or sense RNA of an ankyrin repeat-containing gene blocks chloroplast differentiation in *Arabidopsis*. Plant Cell 4, 1575–1588. <http://dx.doi.org/10.1105/tpc.4.12.1575>.